

RICARDO ALTAMIRANO MARQUEZ

AI Engineer | Agentic Systems & Full-Stack Infrastructure

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SUMMARY

AI Engineer and Full-Stack Engineer building production AI-native systems for domain-heavy operational workflows. Experienced in agent harness engineering, forward-deployed AI solution design, context engineering, RAG, tool/API execution, structured outputs, multimodal analysis, and cloud-deployed applications using Python, FastAPI, React, SQL, Docker, AWS, GCP and Azure.

SKILLS

Programming & Web: Python, TypeScript, JavaScript, Go, SQL, FastAPI, React, REST APIs, WebSockets

AI Systems: Agent harnesses, context engineering, RAG, tool/API execution, MCP-style integrations, multi-agent orchestration, structured outputs, multimodal analysis, LangChain, LangGraph, vector databases

Cloud & Infra: PostgreSQL, MongoDB, Redis, Docker, AWS, Azure, GCP, Terraform, CI/CD, GitHub Actions

Platforms & Integrations: MCP-based tooling, agent infrastructure patterns, permissioned tool use, Microsoft Copilot Studio, Vertex AI, Gemini Enterprise Agent, API integrations

EXPERIENCE

Hydrous Group | AI Engineer / Forward-Deployed AI Engineer

Jan 2025 - Present | hydrousmgmt.com

- Built AI-native systems for water and waste operations, connecting domain workflows with retrieval, tool execution, structured outputs, and production-ready agent workflows.
- Designed agent harnesses coordinating specialized agents for orchestration, regulatory analysis, technical reasoning, document generation, and controlled tool/API execution.
- Applied context engineering patterns to manage domain knowledge, external provider data, technical documents, climate/location context, and agent-specific instructions.
- Worked with startups and clients to define, audit, and implement AI solutions, translating business problems into practical architectures and production-ready workflows.
- Developed multimodal AI and proposal-generation workflows combining documents, images, charts, engineering assets, dynamic tables, process-flow diagrams, and review-ready outputs.

DVAWEB | Software Engineer

Nov 2023 - Jan 2025

- Built and shipped full-stack product features for an intelligent parking platform across frontend, backend services, APIs, databases, and production rollout.
- Developed internal AI tools, including CRM-focused knowledge retrieval and natural-language-to-SQL workflows for non-technical teams.
- Built an enterprise knowledge platform over company documentation and prior projects, integrating proposal and CRM workflows in real time and reducing proposal turnaround time by 78%.
- Contributed to product architecture, debugging, API integrations, performance improvements, and maintainable feature delivery in a production environment.

EDUCATION

Universidad Autonoma de Occidente | B.S. in Software Engineering